

EDUCATION

University of California, Los Angeles
Ph.D. Candidate in Computer Science.
Advisor: Quanquan Gu

Los Angeles, CA, USA
09/2023 - Current

University of California, Los Angeles
M.Sc. in Computer Science

Los Angeles, CA, USA
09/2023 - 12/2025

Tsinghua University
B.E. in Computer Science and Technology

Beijing, China
08/2019 - 07/2023

RESEARCH INTERESTS

Machine Learning, Algorithms and Statistical Limits of (Sequential) Decision Making
Reinforcement Learning and Efficient Methods for Large Language Models

RESEARCH AND WORKING EXPERIENCE

Knowledge Engineering Group, Tsinghua University
Research Assistant, Advisor: Prof. Jie Tang

07/2021 - 01/2022

- Developed P-tuning V2 method (a prefix-tuning-like method) that is comparable with full-finetune universally across model scales and tasks.
- Investigated the performance of P-tuning on neural text retriever.

Knowledge Engineering Group, Tsinghua University
Research Assistant, Advisor: Prof. Juanzi Li

03/2023 - 06/2023

- Developed efficient in-context method for LLM to solve open information retrieval task.
- Investigated LLM's ability in understanding video when visual tools are available.

ByteDance Inc.

06/2024 - 09/2025

Research Scientist Intern, Mentor: Renjie Zheng

- Designed actor-critic reinforcement learning algorithm for LLM post-training
- Developed direct-preference-learning styled offline RL algorithm to enhance reasoning ability of LLMs.
- Applied reinforcement learning methods for training memory agent.

RESEARCH PUBLICATIONS

- [1] **Kaixuan Ji***, Qiwei Di*, Heyang Zhao, Qingyue Zhao, and Quanquan Gu. On the optimal sample complexity of offline multi-armed bandits with KL regularization. In *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- [2] Qingyue Zhao*, **Kaixuan Ji***, Heyang Zhao, and Quanquan Gu. Fast rates for offline contextual bandits with forward-KL regularization under single-policy concentrability. In *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026. **Spotlight Presentation.**
- [3] **Kaixuan Ji***, Qingyue Zhao*, Heyang Zhao*, Qiwei Di, and Quanquan Gu. Near-optimal regret for KL-regularized multi-armed bandits. In *Forty-third International Conference on Machine Learning (ICML)*, 2026.

- [4] Qiwei Di*, **Kaixuan Ji***, Xuheng Li*, Heyang Zhao, and Quanquan Gu. Best-of-majority: Minimax-optimal strategy for pass@k inference scaling. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- [5] Qingyue Zhao*, **Kaixuan Ji***, Heyang Zhao*, Tong Zhang, and Quanquan Gu. Towards a sharp analysis of offline policy learning for f -divergence-regularized contextual bandits. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- [6] **Kaixuan Ji***, Jiafan He*, and Quanquan Gu. Reinforcement learning from human feedback with active queries. *Transactions on Machine Learning Research (TMLR)*, 2025. **Featured Certification**.
- [7] Yue Wu*, Zhiqing Sun*, Huizhuo Yuan*, **Kaixuan Ji**, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- [8] Huizhuo Yuan*, Zixiang Chen*, **Kaixuan Ji***, and Quanquan Gu. Self-play fine-tuning of diffusion models for text-to-image generation. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
- [9] Zixiang Chen*, Yihe Deng*, Huizhuo Yuan*, **Kaixuan Ji**, and Quanquan Gu. Self-play fine-tuning converts weak language models to strong language models. In *Forty-first International Conference on Machine Learning (ICML)*, 2024.
- [10] **Kaixuan Ji***, Qingyue Zhao*, Jiafan He, Weitong Zhang, and Quanquan Gu. Horizon-free reinforcement learning in adversarial linear mixture MDPs. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [11] Weng Lam Tam*, Xiao Liu*, **Kaixuan Ji**, Lilong Xue, Jiahua Liu, Tao Li, Yuxiao Dong, and Jie Tang. Parameter-efficient prompt tuning makes generalized and calibrated neural text retrievers. In *Findings of the Association for Computational Linguistics: EMNLP 2023 (EMNLP-Findings)*, 2023.
- [12] Xiao Liu*, **Kaixuan Ji***, Yicheng Fu*, Weng Lam Tam, Zhengxiao Du, Zhilin Yang, and Jie Tang. P-tuning: Prompt tuning can be comparable to fine-tuning across scales and tasks. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

MANUSCRIPTS UNDER REVIEW

- [1] Kaixuan Ji, Guanlin Liu, Ning Dai, Qingping Yang, Renjie Zheng, Zheng Wu, Chen Dun, Quanquan Gu, and Lin Yan. Enhancing multi-step reasoning abilities of language models through direct q-function optimization. *arXiv preprint arXiv:2410.09302*, 2024.
- [2] Ji Qi, Kaixuan Ji, Xiaozhi Wang, Jifan Yu, Kaisheng Zeng, Lei Hou, Juanzi Li, and Bin Xu. Mastering the task of open information extraction with large language models and consistent reasoning environment. *arXiv preprint arXiv:2310.10590*, 2023.

ACADEMIC SERVICES

Conference Reviewer

- International Conference on Machine Learning (ICML): 2025, 2026
- Annual Conference on Neural Information Processing Systems (NeurIPS): 2024, 2025
- International Conference on Learning Representations (ICLR): 2024, 2025, 2026
- Association for the Advancement of Artificial Intelligence (AAAI): 2026
- International Conference on Artificial Intelligence and Statistics (AISTATS): 2024, 2025, 2026

Journal Reviewer

- IEEE Transactions on Information Theory (TIT)

TEACHING

Teaching Assistant

Fall 2024, Winter 2025, Spring 2025, Fall 2025

CS35L Software Construction, University of California, Los Angeles